σ-ADDITIVE PROBABILITY ON ORTHOMODULAR LATTICES: A SURVEY AND AN OPEN PROBLEM

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ABSTRACT. On a Boolean algebra, a finitely coherent probability extends to a countably additive measure by classical machinery (Carathéodory; Loomis–Sikorski; Stone). On a non-distributive orthomodular lattice (OML) this machinery fails. We survey the problem and isolate a precise open question, splitting it into two axes the Boolean case fuses: the *extension* axis (does a state extend to a charge on the dual?), a Horn–Tarski feasibility question settled negatively for $L(H)$; and the *descent* axis (does the charge concentrate on the physical points?). The combinatorial route fails — $\prod_n \mathrm{MO}_2$ is concrete, σ-complete and non-Boolean yet *segregated* (non-distributivity confined to finite blocks) — because a faithful set representation forces the OML Boolean (Theorem 5.2). Reading "no hidden realisation" as not being a mixture of dispersion-free states makes a relational measure a *contextual* state (Lemma 5.3). The open question: *does a non-Boolean, σ-complete, concrete OML carry a σ-additive contextual state whose contextuality is σ-essential* — witnessed by no finite sub-OML? The finite case is inhabited (Wright's pentagon), so only the σ-essential refinement remains, a compactness failure of CE type. Recent work on Dynkin-systems (Derr–Williamson [9]) settles it *negatively under a Polish/Borel-regularity hypothesis*, leaving the residue — whether a witness can be non-Polish-representable — a descriptive-set-theory question (Remark 5.1).

1. Introduction

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*:

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_n A_n = \varnothing \quad \Longrightarrow \quad \mu(A_n) \to 0.$$

De Finetti [8] held that only finite additivity is empirically grounded; Kolmogorov [27] adopted σ-additivity as an axiom.

The same tension recurs once observations need not be jointly performable. An observation *context* — a set of measurements performable together — is Boolean; *pasting* contexts along their shared observations (the horizontal-sum / partial-Boolean-algebra construction [23]) returns:

$$\begin{aligned} \text{contexts mutually compatible} &\Rightarrow \text{Boolean algebra,} \\ \text{incompatible} &\Rightarrow \text{non-distributive OML.} \end{aligned}$$

The pasting of incompatible Boolean blocks is a non-distributive OML (Kalmbach [23]), and its failure to remain Boolean is exactly the contextuality obstruction of the partial-Boolean-algebra / sheaf account (Abramsky–Brandenburger [1]; Budroni–Morchio [2]). The OML is thus *forced* by the context structure, not postulated; for non-quantum examples the incompatibility must be operationally imposed, since classical observables on a common space jointly distribute (Kolmogorov) and stay Boolean. The closed subspaces of a Hilbert space H form the prototype $L(H)$, not the premise.

On a non-distributive OML, Carathéodory's outer measure does not apply. For $L(H)$ it is rescued by Gleason's theorem [16], with no general-OML analogue known. Non-distributivity *per se* does not

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remove the two-valued homomorphisms the Boolean engine needs (MO_3 keeps them), but it removes the *guarantee*; for $L(H)$, $\dim \geq 3$, they fail outright (Kochen–Specker [26]). Whether enough survive is a question of *concreteness*, not non-distributivity: an OML is *concrete* (Definition 2.7) when it carries an order-determining set of two-valued states (Gudder [18]), and this is the qualifier on which the open problem turns.

The organising split. The question splits into two axes the Boolean case fuses: *extension* (does a state extend to a charge on $S_0(A)$? — settled, none does on $L(H)$) and *descent* (does the charge concentrate on the physical points? — open).

The relational standpoint. Following De Finetti [8] (probability without a presupposed sample space) and Birkhoff–von Neumann (propositions as a lattice, not subsets of a phase space), we start from the propositions and ask what their order alone determines — the measure-theoretic counterpart of pointless topology. The dual is then the McDonald–Bimbó space of *filters* (§2.3), and a positive answer would be a σ-additive probability theory on a non-distributive lattice, the non-Boolean analogue of localic measure theory (made precise in §5 as a *contextual* state; cf. [29]).

2. Preliminaries

We collect the lattice-theoretic apparatus; all notions are standard.

2.1. **Orthomodular lattices.**

Definition 2.1 (Orthocomplemented lattice)**.** A *bounded lattice* is a partially ordered set in which every pair a, b has a join $a \vee b$ and meet $a \wedge b$, with greatest element $\mathbf{1}$ and least element $\mathbf{0}$. An *orthocomplementation* is a map $a \mapsto a^\perp$ satisfying, for all a, b:

(i) $a \wedge a^\perp = \mathbf{0}$ and $a \vee a^\perp = \mathbf{1}$ (complement),
(ii) $a^{\perp\perp} = a$ (involution),
(iii) $a \leq b \Rightarrow b^\perp \leq a^\perp$ (order-reversing).

Definition 2.2 (Incomparable; meet-zero; orthogonal)**.** For elements a, b of an orthocomplemented lattice, three relations of increasing strength:

- *incomparable*: $a \not\leq b$ and $b \not\leq a$ — neither entails the other (an order condition);
- *meet-zero*: $a \wedge b = \mathbf{0}$ — no common refinement;
- *orthogonal*, written $a \perp b$: $a \leq b^\perp$ (equivalently $b \leq a^\perp$) — one entails the other's negation, *decisively incompatible.*

Each implies the one above it ($a \perp b \Rightarrow a \wedge b = \mathbf{0} \Rightarrow$ incomparable, for $a, b \notin \{\mathbf{0}, \mathbf{1}\}$). For two *distinct atoms*, incomparable and meet-zero coincide — an atom has only $\mathbf{0}$ below it, so $a \wedge b \in \{\mathbf{0}, a\}$, and $a \wedge b = a$ would force $a \leq b$; orthogonality remains strictly stronger.

Remark 2.3 (Meet-zero versus orthogonal). The implication $a \perp b \Rightarrow a \wedge b = \mathbf{0}$ reverses in every Boolean algebra, but fails once the lattice is non-distributive: there meet-zero is strictly weaker than orthogonality. The distinctions developed below all turn on this separation.

Definition 2.4 (Orthomodular lattice)**.** An orthocomplemented lattice A is *orthomodular* if

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^\perp \wedge b). \tag{1}$$

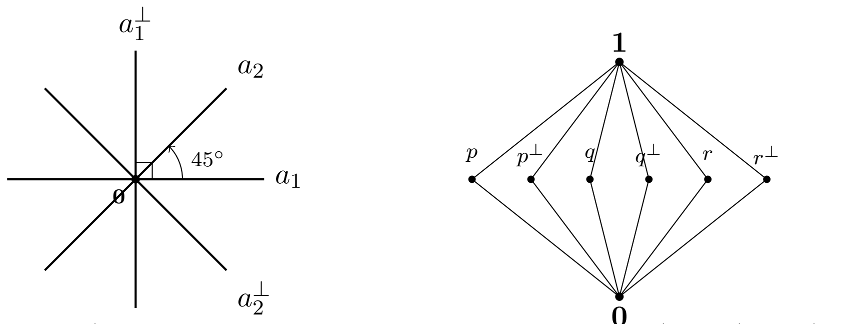
A *Boolean algebra* is the special case where $\wedge$ distributes over $\vee$; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is $L(H)$, the closed subspaces of a Hilbert space with $a^\perp$ the orthogonal complement.

Definition 2.5 (Completeness notions)**.** Two countable-completeness conditions on an OML A are distinguished throughout, the second strictly weaker. Here $\bigvee_n a_n$ denotes the *join* (least upper bound) of $\{a_n\}$ in the order of A, unique when it exists; "$\exists \bigvee_n a_n \in A$" asserts that it does.

- A is *σ-complete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A : \quad \exists \bigvee_n a_n \in A$$
(equivalently every countable family has a meet, by $\bigwedge_n a_n = (\bigvee_n a_n^{\perp})^{\perp}$);
- A is *σ-orthocomplete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A \ \big(a_i \perp a_j \text{ for } i \neq j\big) : \quad \exists \bigvee_n a_n \in A.$$

The orthogonal families are a subclass of all countable families, so σ-completeness quantifies over a strictly larger domain and is the stronger hypothesis: σ-completeness $\Rightarrow$ σ-orthocompleteness. Only the weaker condition is needed below. For a state s, σ-additivity is the requirement that
$$s\Big(\bigvee_n a_n\Big) = \sum_n s(a_n)$$
for every countable pairwise-orthogonal family $\{a_n\}$. Its left side invokes $\bigvee_n a_n$ *only* for orthogonal families — so σ-orthocompleteness is exactly the hypothesis that makes it well-posed. Full σ-completeness is the stronger condition manipulated by the Loomis–Sikorski representation (§2.4).

Example 2.6 (MO$_n$, and the gap made concrete)**.** MO$_n$ is the lattice with $\mathbf{0}$, $\mathbf{1}$, and n complementary pairs of incomparable atoms, all sharing only $\mathbf{0}$ and $\mathbf{1}$. The geometric model is n lines through the origin: each line a is orthogonal to its perpendicular $a^{\perp}$, while two atoms from distinct pairs are non-perpendicular lines. It is orthomodular and non-distributive ($n \geq 2$): two atoms drawn from *distinct* complementary pairs have
$$a \wedge b = \mathbf{0} \quad \text{(meet only at the origin)} \qquad \text{yet} \qquad a \not\perp b,$$
so meet-zero $\neq$ orthogonal already here. MO$_3$ is the smallest non-distributive OML.



$a_1 \perp a_1^{\perp}$ (right angle),
$a_1 \not\perp a_2$ but $a_1 \wedge a_2 = \mathbf{0}$

complementary pairs $(p,p^{\perp}),(q,q^{\perp}),(r,r^{\perp})$;
any two non-paired atoms meet at $\mathbf{0}$, join at $\mathbf{1}$

FIGURE 1. MO$_n$ illustrated. *Left:* the geometric model for MO$_2$ — four atoms as lines through the origin in $\mathbb{R}^2$. The pair $a_1, a_1^{\perp}$ meets at a right angle (orthogonal), whereas a_1, a_2 meet at 45°: their lattice meet is still $\mathbf{0}$ (only the origin in common), yet they are *not* orthogonal — the meet-zero $\neq$ orthogonal gap. *Right:* the Hasse diagram of MO$_3$, the smallest non-distributive OML: three complementary pairs of incomparable atoms between $\mathbf{0}$ and $\mathbf{1}$.

Definition 2.7 (OMP; order-determining states; concrete logic)**.** An *orthomodular poset* (OMP) weakens Definition 2.4: joins $a \vee b$ are required only for *orthogonal* pairs. A set S of states on A is *order-determining* if it separates the order, i.e.
$$\big(s(a) \leq s(b) \text{ for all } s \in S\big) \implies a \leq b.$$
An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into $(\mathcal{P}(X), \subseteq, X \setminus (-))$ for some set X, preserving existing orthogonal joins — i.e. its propositions are modelled as actual *sets*

of outcomes. Concreteness is equivalent to admitting an order-determining set of *two-valued* states (Gudder [18]), and it is the property that separates the open frontier from the settled cases (§5).

Remark 2.8 (Concreteness does not close the gap). A set-representable OMP needs only closure under complement and *disjoint* union [4]; intersection-closure would force a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L-member inside $A \cap B$, so

$$\underbrace{A \cap B = \varnothing}_{\text{orthogonal}} \implies \underbrace{a \wedge b = \mathbf{0}}_{\text{meet-zero}}, \qquad \text{but not conversely.}$$

Concreteness does not force the converse: MO_3 is itself concrete (Definition 2.7; Gudder [18]), so the paradigmatic meet-zero-$\neq$-orthogonal example is already a concrete logic. What closes the gap is a richness/atomicity condition, not concreteness:

$$a \wedge b = \mathbf{0} \;\Rightarrow\; a \perp b,$$

equivalently: every nonempty $A \cap B$ contains a nonzero L-member — the abstract form being Tkadlec's *Boolean orthoposet* condition [39] (which does *not* mean Boolean *algebra*).

2.2. The state/meet-additive/charge ladder.

Definition 2.9 (State; meet-additive state; charge-extendible)**.** Let A be an OML. A *state* is a map $s : A \to [0,1]$ with $s(\mathbf{1}) = 1$ that is additive on *orthogonal* pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$; that is, a probability assignment whose additivity is guaranteed only on decisively incompatible propositions. Three strengthenings, differing in which pairs additivity is demanded on and strictly increasing in general (Example 2.10 on MO_3):

(1) *State*: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$.
(2) *Meet-additive*: $s(a \vee b) = s(a) + s(b)$ whenever $a \wedge b = \mathbf{0}$ (binary, strictly stronger — it reaches the gap pairs of Definition 2.2). We avoid the word "valuation" here: in lattice theory it denotes the modular function $v(a) + v(b) = v(a \wedge b) + v(a \vee b)$, a different condition.
(3) *Charge-extendible*: $\sum_i s(a_i) \le 1$ for every pairwise-meet-zero family $\{a_i\}$ ($a_i \wedge a_j = \mathbf{0}$, $i \neq j$; strictly stronger again). This is the condition necessary for extension to a charge on the dual space (Definition 2.13) and the *extension* condition of §3.

Example 2.10 (The ladder is strict)**.** On MO_3 with complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$, the maximally mixed $s \equiv \frac{1}{2}$ is meet-additive (clears (2)) yet fails (3) at $\{p, q, r\}$: $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So no state on MO_3 is charge-extendible.

Definition 2.11 (Normal and singular states on $L(H)$)**.** On $A = L(H)$ a state s is

- *normal* if it is completely additive on every orthogonal family of projections,
$$s\Big(\bigvee_{i \in I} p_i\Big) = \sum_{i \in I} s(p_i) \quad \text{for all (not only countable) orthogonal } \{p_i\},$$
equivalently $s(\cdot) = \mathrm{tr}(\rho\, \cdot)$ for a density operator ρ (Gleason [16], $\dim \geq 3$);
- *singular* (purely) if
$$s(e) = 0 \quad \text{for every finite-rank projection } e.$$

Every state decomposes into a normal and a singular part (Takesaki); the two classes are exactly those distinguished by the arguments of §3.

2.3. The McDonald–Bimbó dual space.

Definition 2.12 (Dual space; representation map)**.** The McDonald–Bimbó duality [30] assigns to an OML A a compact space

$$S_0(A) = \big(F(A), \subseteq, \perp_A, P(A), T\big),$$

where $F(A)$ is the set of *filters* of A, ordered by $\subseteq$ and carrying an orthogonality relation $\perp_A$ and a Stone topology T, and $P(A) = \{ \uparrow p : p \in A \} \subseteq F(A)$, with $\uparrow p = \{a \in A : a \geq p\}$, is the set of *principal* filters — the "physical points," the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For $a \in A$ set $h(a) = \{x \in F(A) : a \in x\}$; then h is an OML isomorphism onto the $\perp$-stable clopens $\mathrm{CO}(S_0(A))^\dagger$. Define $\mu(h(a)) = s(a)$. Here $h(a)$ is the set of points at which a holds and μ is the state transported onto those sets; the open problem is whether μ is preserved under countable operations on them.

Definition 2.13 (Charge)**.** A *charge* on $S_0(A)$ is a map $\nu : \mathrm{CO}(S_0(A)) \to [0,1]$ on the *full* Boolean algebra of clopen subsets (not merely the $\perp$-stable ones in the image of h), with

$$\nu(S_0(A)) = 1, \qquad \nu(U \cup V) = \nu(U) + \nu(V) \ \text{ whenever } U \cap V = \varnothing.$$

A state s *extends to a charge* if some such ν satisfies $\nu(h(a)) = s(a)$ for all $a \in A$. Because h preserves meets, a pairwise-meet-zero family in A maps to disjoint clopens, so a charge necessarily satisfies $\sum_i s(a_i) \leq 1$ over such families — the charge-extendibility condition of Definition 2.9(3).

Remark 2.14 (The duality is finitary)*.* The representation map h satisfies

$$h\Big(\bigvee_{i\leq n} a_i\Big) = \Big(\bigcup_{i\leq n} h(a_i)\Big)^{\perp\perp} \quad \text{(finite)},$$
$$h\Big(\bigvee_{n} a_n\Big) \neq \Big(\bigcup_{n} h(a_n)\Big)^{\perp\perp} \quad \text{(countable)}:$$

finitary OML homomorphisms need not preserve infinite suprema. This is the central obstruction the open problem of §5 confronts, and it is self-dual: since $a \mapsto a^\perp$ is an order anti-isomorphism, $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$, so the failure for countable meets is the De Morgan dual of the failure for countable joins. The rival duality of Cannon–Döring [5] is also finitary and carries no measure; we use McDonald–Bimbó because it carries $P(A)$ as explicit structure.

2.4. **The Boolean engine: Stone duality.** In the Boolean case the descent equivalence

$$\sigma\text{-additive} \quad \Longleftrightarrow \quad \text{the measure concentrates on the physical points}$$

is proved through *Stone duality*, in four steps [29]:

- (S1) a Boolean event algebra B embeds in the clopen algebra $\mathrm{CO}(\mathrm{St}\, B)$ of its Stone space;
- (S2) a charge on B extends to a Baire measure on $\mathrm{St}\, B$ (Carathéodory);
- (S3) σ-additivity $\Longleftrightarrow$ the measure vanishes on meagre Baire sets (Rao–Rao [36]);
- (S4) meagre-null $\Longleftrightarrow$ the measure lives on the ultrafilters closed under countable intersection = the realised points.

Steps (S3)–(S4) are the measure-theoretic face of the *Loomis–Sikorski* representation,

$$\sigma\text{-complete Boolean algebra} \ \cong \ (\sigma\text{-tribe of sets}) \,/\, (\sigma\text{-ideal}),$$

the countable-join bookkeeping that drives the equivalence. The contrast with the OML case is exact: the Boolean engine (Stone duality together with the Loomis–Sikorski representation) has *no* known OML analogue (§4.3). Supplying such an engine, or proving none exists, is the open problem of §5.

Remark 2.15 (Why the bridge fails: meet-closed but not join-prime)*.* A "realised" point x carries two conditions, which Rao–Rao's "vanishes on meagre sets" makes coincide in the Boolean case:

- (M) closed under countable meets;
- (J) $x \ni \bigvee_n a_n \Rightarrow x \ni a_n$ for some n (off the join-defect).

Non-distributivity separates them. For a principal filter $x = \uparrow p$, condition (M) always holds, whereas (J) can fail: $p \le \bigvee_n a_n$ does not force $p \le a_n$ for some n. In $L(H)$ this is witnessed by $a_n = \mathrm{span}(e_n)$ and $p = \mathrm{span}(e_1 + e_2)$:

$$p \ \le \ \bigvee_n a_n = H, \qquad \text{yet} \qquad p \ \not\le \ a_n \ \text{ for every } n.$$

So $\uparrow p$ is a realised point lying inside the join-defect $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$. Topologically $\uparrow p$ is *isolated* in $S_0(L(H))$, hence:

$$D \text{ non-meagre (OML)} \qquad \text{vs.} \qquad D \text{ nowhere dense (Boolean).}$$

This is the finitary obstruction of Remark 2.14 on the dual: the category route (clopens modulo the meagre ideal) fails on the same D as the measure route, and reduces to the MacNeille completion of §4.3(ii).

3. Extension and descent

The central question — *when does a state s on an OML A extend to a σ-additive measure on $S_0(A)$?* — splits into two *independent* axes that the Boolean case fuses.

Extension.:
Does s extend to a finitely additive charge on the *full* Boolean clopen algebra of $S_0(A)$? This is governed by the meet-zero/orthogonal gap (§2.1) and is *settled*: it is a Horn–Tarski/Pitowsky feasibility question, a decidable linear program in the finite case, and can fail for every state.

Descent.:
Once extended, does the charge concentrate on the physical points $P(A)$? This is governed by σ-additivity — it needs the missing engine (§2.4) — and is *open*.

The extension axis is the abstract form of a familiar question: extending a finitely-additive assignment on incompatible propositions to a single joint distribution is exactly the feasibility problem behind the Bell inequalities (Fine [13]: a Bell model exists iff the correlations admit a joint distribution; Pitowsky [33]: the Bell inequalities are the facets of a correlation polytope), and Bell and Kochen–Specker non-classicality are jointly an extension problem for partial Boolean algebras (Budroni–Morchio [2], with the necessary-and-sufficient conditions of Horn–Tarski [22]). It is also the marginal problem of probability theory — extending consistent marginals to a joint law (Kellerer [24]) — and the coherence/extendability question of imprecise probability (Walley [42]); the equivalence of these formulations to the quantum-logical one is made precise by Derr–Williamson [9]. What is distinctive here is only that the extension is sought against the dual $S_0(A)$ rather than within a fixed scenario; the obstruction is the same finitary one, and it is settled. Two observations pin it down.

Proposition 3.1 (Finite case)**.** *For finite A the extension condition is a decidable linear program (Horn–Tarski* [22] */ Pitowsky correlation-polytope feasibility* [33]*; Budroni–Morchio* [2] *for the partial-Boolean-algebra formulation); since h preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy $\sum_i s(a_i) \le 1$ over pairwise-meet-zero families. It can fail for* every *state (*MO_3*, Example 2.10).*

Proposition 3.2 (Infinite $L(H)$, all states)**.** *Let* $\dim H = \infty$*.* No *state on $L(H)$ — normal or singular — extends to a charge on $S_0(L(H))$.*

Proof sketch. Embed MO_2 in $L(H)$ with *infinite-dimensional* atoms: writing $H = H_0 \otimes \mathbb{C}^2$ ($\dim H_0 = \infty$), the four subspaces $H_0 \otimes \mathbb{C}e$, $H_0 \otimes \mathbb{C}f$, $H_0 \otimes \mathbb{C}(e+f)$, $H_0 \otimes \mathbb{C}(e-f)$ are pairwise meet-zero with $a_i \vee a_i^{\perp} = H$. Two counts of $\sum_i s(a_i)$ now collide. Orthoadditivity with $s(\mathbf{1}) = 1$ gives $s(a_i) + s(a_i^{\perp}) = 1$, so for every state

$$\sum_i s(a_i) = 2.$$

The meet-preserving h sends this pairwise-meet-zero family to disjoint sets, so any charge satisfies

$$\sum_i s(a_i) \leq 1.$$

The two are incompatible for every state, so none extends. The construction is finite ($n = 4$), so σ-completeness plays no role; the infinite-dimensional atoms are what carry the obstruction past the finite-rank states to the singular ones (where every finite-rank projection has measure zero), closing the axis completely. □

4. What is known

The extension axis is settled — on $L(H)$ no state extends (Proposition 3.2); the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. Three things bear on that:

1. the Boolean benchmark such an analogue must match (§4.1);
2. the partial OML results, positive and obstructive (§4.2);
3. the routes already known to be blocked (§4.3).

4.1. **The Boolean benchmark.** The conditions an OML engine would have to reproduce. The Kelley–Vladimirov–Pták characterisation (Fremlin [14], Thm 391D) is complete: a Boolean algebra carries a strictly positive σ-additive measure iff it satisfies all three conditions below — each of which loses its meaning off the distributive world.

Boolean (KVP) condition	**OML analogue**
Dedekind σ-completeness	exists (σ-orthocompleteness)
weak (σ, ∞)-distributivity	none — relies on $\bigwedge$ distributing over $\bigvee$
chargeability	none — no structural characterisation on OMLs

4.2. **Known results for general OMLs. Positive** (require structure rich enough to approximate Hilbert-space geometry):

- Gleason [16] — on $L(H)$ for a *separable* Hilbert space of dim ≥ 3, every countably additive measure on the projection lattice has the trace form $s(\cdot) = \mathrm{tr}(\rho\,\cdot)$: σ-additivity *on the lattice*.
- Bunce–Wright [3] — the projection-lattice (JBW) form; Chetcuti–Dvurečenskij [6].

Obstructive (Boolean machinery cannot be transplanted):

- Pták–Pulmannová [35] — a *finitary* condition (*subadditive*-unitality $s(a \vee b) \leq s(a) + s(b)$, no countable joins) already collapses the *lattice* to Boolean (their Thm 1); the OM*poset* version fails (Müller's set-representable counterexample [31]).
- Navara [32] — regularity does not force σ-additivity; with Rüttimann's decomposition of states into completely additive and purely finitely additive parts [37], this shows the finitely-additive/σ-additive gap is structurally serious on quantum logics, not a routine strengthening — part of why the open problem is hard. Pták [34] — exotic state spaces via Greechie pasting.

4.3. **Three routes to a σ-OML engine.** The descent argument needs an OML analogue of the Boolean engine — a σ-Stone duality, equivalently a Loomis–Sikorski representation. Routes (i),(ii) are closed by *theorem*; route (iii) is open in the strong sense (no construction known, none ruled out), and is the Open Problem of §5.

(i) **RDP / effect-algebra.** Loomis–Sikorski holds for σ-MV algebras [11] and for monotone σ-complete effect algebras *with* the Riesz Decomposition Property [12]; but a lattice effect algebra has RDP iff it is MV, and OML $\cap$ MV = Boolean (Kalmbach [23]): OML+RDP $\iff$ Boolean. The *sharp-skeleton* evasion (represent the OML as the sharp elements of an ambient

RDP effect algebra) also closes: in an RDP effect algebra the sharp elements coincide with the centre, hence are Boolean (Jenča 2001, Cor. 4.3, via Greechie–Foulis–Pulmannová [17]).

(ii) **MacNeille completion.** The MacNeille completion of an OML need not be orthomodular (Harding [19]), so one cannot complete to absorb countable joins; the topological version (clopens of $S_0(A)$ modulo the meagre ideal) is the same completion and collapses non-distributivity (Remark 2.15).

(iii) **σ-Stone duality.** None is known, none ruled out: the existing dualities are finitary (McDonald–Bimbó, Remark 2.14) or equational with no set/tribe representation (Freytes [15]), but no theorem forbids a countable-join-preserving one. This is route (c) of the faithfulness trichotomy (§5.1): a representation faithful on countable *orthogonal* joins without forcing distributivity.

4.4. **Prior art on point-free quantum probability.** Every existing point-free or directed-context construction fails at least one of the two requirements — σ-additivity and a natively non-distributive structure — as tabulated below.

Gleason tradition: Varadarajan [40], Kalmbach [23], Pták–Pulmannová [35]: σ-additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

Döring 2009: [10] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

Bohrification (HLS): [20, 21] point-free and the internal valuation *is* σ-additive (Scott-continuous) — but factors through an internally *distributive* locale, σ-additive only per commutative context. Non-distributivity is tamed, not carried natively.

Localic valuations: Vickers [41], Simpson [38], Coquand–Spitters [7]: point-free σ-additive, but on *frames* — distributive by definition.

OML Stone dualities: McDonald–Bimbó [30] and Cannon–Döring [5]: purely structural, no measure, no σ-version.

The open case is their conjunction, met by no existing construction:

$$\boxed{\text{point-free} \times \sigma\text{-additive} \times \text{natively non-distributive} \times \text{directed-system-built.}}$$

Remark 4.1 (The boundary of the negative results). Two qualifications delimit what is ruled out.

(i): **No impossibility theorem covers the concrete class.** The known no-go results are finite or two-valued, and the one positive occupant is non-concrete:

$$L(H)\ (\dim\!\geq\!3): \quad \text{no two-valued states at all (Kochen–Specker [26]),}$$

a fortiori none order-determining. Concreteness is the qualifier on which the open frontier turns.

(ii): **No σ-Loomis–Sikorski exists** (route (c), §5.1): the dualities take infinite joins as a closure, not a union (Remark 2.14). Not to be overstated, though: Gudder's concrete logics [18] *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a σ-Loomis–Sikorski theorem.

4.5. **The topos approach.** The topos / Bohrification programme is the one machinery supplying a point-free *and* σ-additive valuation on a quantum logic, so it is the natural candidate — but it takes route (b) of the trichotomy (§5.1): in both principal constructions the valuation lives on a *distributive* object, its countable additivity routed through per-context Boolean blocks, hence sidesteps rather than settles route (c).

	Bohrification (HLS)	**Döring–Isham**
measure on	internal Gelfand spectrum $\Sigma(A)$, a compact regular frame	clopen subobjects $\mathrm{Sub}_{cl}(\Sigma)$, a Heyting algebra
why distributive	a frame: $\bigvee$ distributes [21, Def. 6.11]	a locale, not a σ-algebra [10]
how Proj enters	Thm 6.19 bijection; the measure restricts to a σ-Boolean morphism on each Boolean sublattice (Def. 6.17(a))	daseinisation, which "cannot preserve both [joins and meets]" [43]
additivity left	block-by-block, glued by naturality; never crosses a non-commuting join	only *finitely* additive; σ per-block, normal states only

Two route-(b) details specific to the topos case: HLS confines all additivity inside Boolean contexts (the antithesis of σ-essential contextuality), and its non-distributive object $\mathrm{Proj}(A)$ is the non-concrete Gleason / $L(H)$ lattice (Rmk 4.1), not the concrete class in question — so of the four requirements (point-free, σ-additive, natively non-distributive, directed-system-built) it misses exactly native non-distributivity.

5. The open problem

Open Problem. Does there exist a concrete, σ-complete, non-Boolean OML L and a σ-additive state w on L such that

$$\begin{aligned} &\textit{(contextual)} && w \notin \overline{\mathrm{conv}}\,(S^{\sigma}_{\mathrm{df}}); \\ &\textit{(σ-essential)} && w|_F \in \overline{\mathrm{conv}}\,(S_{\mathrm{df}}(F)) \text{ for every finite sub-OML } F \subseteq L\,? \end{aligned}$$

That is, w has no global hidden joint over the dispersion-free states, yet its restriction to every finite sub-OML does. Equivalently: can one prove no such (L, w) exists?

Remark 5.1 (Relation to Derr–Williamson 2023)*.* The problem above is not plainly open: it must be read against Derr–Williamson [9], who study coherent probabilities on *pre-Dynkin-systems* (concrete orthomodular posets) and *Dynkin-systems* (concrete σ-complete OMLs) from the imprecise-probability side. Three distinct levels of statement are involved, and should not be conflated:

(a) a finite-additive extendability criterion (their Theorem 4.5, a Horn–Tarski feasibility condition) matching "non-contextual on every finite sub-OML";
(b) a *countably additive* extension to the ambient Borel σ-algebra (their Theorem D.6), valid *under the hypotheses* that the OML is realised as a Dynkin-system D_σ of Borel sets of a *Polish* space Ω with $\sigma(D_\sigma) = \mathrm{Borel}(\Omega)$ and inner-regular σ-blocks — a result they obtain by specialising Maharam's extension theorem [28, Thm 8.1] (Hausdorff base, inner-regular measures); and
(c) the inference from (b) to non-contextuality of w, below, which is the survey's own step.

The link to the present formulation is one-directional but in the direction that matters. A point $\omega \in \Omega$ gives the evaluation δ_ω, which is a two-valued and (since $A_n \downarrow \varnothing \Rightarrow \omega \notin A_n$ eventually) σ-additive dispersion-free state; a countably additive ν on $\mathrm{Borel}(\Omega)$ with $\nu|_{D_\sigma} = w$ exhibits w as a barycentre,

$$w(A) = \int_\Omega \delta_\omega(A)\, d\nu(\omega) \quad \Longrightarrow \quad w \in \overline{\mathrm{conv}}\,(S^{\sigma}_{\mathrm{df}}).$$

Thus

$$\sigma\text{-extendable} \implies \text{non-contextual},$$

and through Theorem D.6 the implication "finitely non-contextual everywhere $\Rightarrow$ globally non-contextual" holds *whenever the OML admits such a Polish Borel realisation*: under that hypothesis the σ-essential cell is empty, and no witness exists. (The converse, non-contextual $\Rightarrow$ σ-extendable, does not hold — the dispersion-free states properly contain the point evaluations δ_ω — but it is not needed for the negative conclusion.)

The residue is then a single condition, the representability the whole problem turns on: can a σ-essential witness exist that admits *no* Polish Borel realisation (S^σ_{df} non-Polish / analytic-but-not-Borel)? This is a descriptive-set-theory question, equivalent to the absence of a σ-Loomis–Sikorski representation (§4.3); the off-centre/non-band constraints of §6 are its lattice-side counterpart. One caution concerns the σ-version of the (A)/(B) point-space equivocation (Remark 5.6): Derr–Williamson's Polish hypothesis is on the underlying *point set* Ω, and obtaining such an Ω for an abstract concrete σ-complete OML is itself the missing engine, not a free input.

Here "relational" means *no hidden realisation*: the measure is not recoverable from an auxiliary space of definite states behind the propositions. Three point-spaces must be kept distinct: the *canonical* dual $P(A) = \{\uparrow p\}$ (one principal filter per element, §2.3); the *dispersion-free states* (the hidden-variable points); and the *external (A)-points* (Gleason's Hilbert rays, Remark 5.6). $P(A)$ is *permitted*; the objection is only to a measure built on an illicitly introduced space — the (A)-points, equivalently a global hidden-variable joint over the dispersion-free states. By Lemma 5.3 a state has a hidden realisation iff it is a mixture of dispersion-free states, so a relational measure is precisely a *contextual* state. This is a condition on the dispersion-free states, not on $P(A)$: a witness is point-rich on $P(A)$ yet has no global dispersion-free joint.

5.1. **Reduction to contextuality.**
The universal floor. The naive route — realise the lattice as honest sets so that a measure becomes a measure on a space — is closed for every non-Boolean OML.

Theorem 5.2 (Universal floor)**.** *A* faithful set representation *of an OML L — an injective $h : L \to \mathcal{P}(\Omega)$ with $h(a \wedge b) = h(a) \cap h(b)$, $h(a \vee b) = h(a) \cup h(b)$, $h(a^\perp) = \Omega \setminus h(a)$ (a σ-tribe / Loomis–Sikorski representation when σ-joins are preserved) — exists only if L is Boolean.*

Proof sketch. Set operations $\cap, \cup, ^c$ on $\mathcal{P}(\Omega)$ distribute, so the image $h(L)$ is a distributive sublattice; h injective and lattice-preserving makes L isomorphic to a distributive ortholattice, i.e. Boolean. ☐

The universal floor (Theorem 5.2) fixes *where* a σ-engine may relinquish faithfulness, a trichotomy organising §§4.3–4.4: **(a)** all $\wedge, \vee$ as $\cap, \cup$ forces Boolean (Theorem 5.2), and is *excluded*; **(b)** all joins but in a *distributive* ambient object (frame/topos) is σ-additive only per Boolean context (§4.5), and *sidesteps* rather than settles the question; **(c)** only countable *orthogonal* joins, faithful to order and $\perp$ but not to the distributivity-forcing meets — a concrete logic, exactly what σ-additivity invokes — is *the remaining open route* (§4.3(iii)), elaborated next. Route (c) is the candidate for the missing σ-duality — the contravariant equivalence one would want, between concrete σ-complete OMLs and a category of σ-spaces whose points are the dispersion-free states. The universal floor fixes three features of the right-side object:

- non-distributivity cannot live in the sets (faithful $\wedge, \vee \mapsto \cap, \cup$ forces Boolean, Theorem 5.2); it sits instead in the meet being a sub-intersection under the Gudder/concrete-logic functor (orthogonal joins as disjoint unions, set-complement);
- the descent space is S^σ_{df}, *not* the principal points $P(A)$ (which are permitted; Remark 5.6);
- the field of sets is σ-generated, not a Stone clopen algebra.

Under that functor the *representation* is settled: a concrete σ-Dynkin-system does not collapse to Boolean (Remark 2.8; $\prod_n \mathrm{MO}_2$ inhabits it). But this is a *definitional pin*, not a theorem — fixing the target to σ-Dynkin-systems makes the representation hold by the chosen functor. The entire weight therefore rests on the one remaining clause, the *measure descent*: whether a σ-additive state

is a mixture of the $S^{\sigma}_{\rm df}$, off the regular-realisable case (Remark 5.1). So a concrete OML has an order-determining family of dispersion-free states, and the one open question is whether every state is a mixture of them.
The reduction. The following equivalence reduces "no hidden realisation" to a spanning condition on the dispersion-free states. Its content is not new — for a Bell scenario it is Fine's theorem [13] (a joint distribution exists iff a deterministic hidden-variable model does), and the convex-geometric form is Pitowsky's correlation-polytope picture [33]; the formulation for a general (concrete) OML, with dispersion-free states as the deterministic vertices, is the natural lift, in the lineage of the sheaf-theoretic account of contextuality [1].

Lemma 5.3 (Relational = contextual)**.** *Let L be a concrete OML with dispersion-free state space $S_{\rm df}$, and w a state on L. Then w admits a hidden realisation — a probability measure on a space of global two-valued valuations of L inducing w — if and only if $w \in \overline{\rm conv}\,(S_{\rm df})$. We call w* contextual *when it is not.*

Proof sketch. A global two-valued valuation of L is exactly a dispersion-free state, so a hidden realisation is a probability measure μ on $S_{\rm df}$ with $w(a) = \int \omega(a)\, d\mu(\omega)$ for all a — i.e. w is the barycentre of μ. The barycentres of probability measures on the (weak-$*$ compact) set $S_{\rm df}$ are exactly $\overline{\rm conv}(S_{\rm df})$. The handle is *spanning*, not simplicity: non-uniqueness of μ is irrelevant; only its *existence* matters. □

The object sought is therefore a concrete σ-complete OML with a σ-additive state outside the dispersion-free hull (Figures 2 and 3).

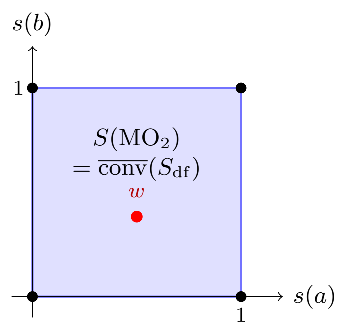


FIGURE 2. MO_2: the state space is the square $[0,1]^2$ (a state is fixed by $(s(a), s(b))$), and the four dispersion-free states are its corners, so $\overline{\rm conv}(S_{\rm df})$ is the whole square. Every state w is a mixture of corners — none is contextual. Thus non-distributivity alone does not produce contextuality (contrast Figure 3): it requires a loop, not merely incompatible elements.

Why σ-essential. The finite version of this is already inhabited. Wright's pentagon [44] is a finite, concrete OML — a Greechie loop of five three-atom blocks, hence a lattice — with an order-determining family of dispersion-free states, yet it carries a non-classical $\{0, \frac{1}{2}\}$-valued state outside their closed convex hull. The reason is a vertex-sum collision. The five atoms form a 5-cycle in which consecutive atoms share a block, so a dispersion-free state takes at most one of each adjacent pair to 1, hence at most 2 of the 5; its hull therefore satisfies

$$\textstyle\sum_{i=1}^{5} \omega(a_i) \;\le\; 2.$$

The $\{0, \frac{1}{2}\}$ state — legitimate, each block summing to 1 — instead has

$$\textstyle\sum_{i=1}^{5} \omega(a_i) \;=\; \frac{5}{2},$$

so it lies outside the hull (this is the KCBS inequality [25]). Thus "concrete OML with a contextual state" is a 1978 fact, and the general spanning statement is false.

What Wright does *not* supply is contextuality that requires the countable structure: the pentagon's is witnessed by a finite sub-OML, namely itself. A σ-essential witness must therefore be contextual in the σ-complete whole but in no finite sub-OML. This is not a device to evade Wright: "every finite piece has a global section, the countable whole has none" is a *compactness failure* of exactly the type by which countable additivity is not first-order axiomatizable over finitely additive coherence (Remark 5.4).

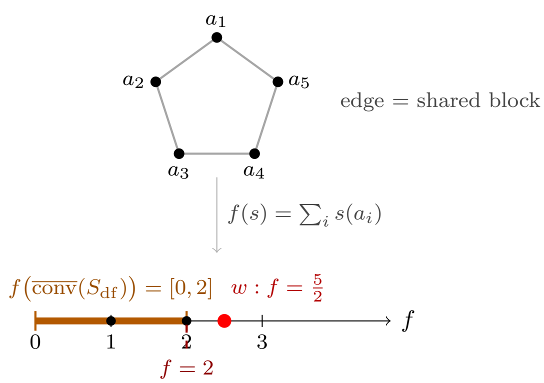


FIGURE 3. Wright's pentagon (a Greechie 5-loop; left), and the same hull-membership question read off a separating functional (right). The state space is 5-dimensional and undrawable, but $f(s) = \sum_i s(a_i)$ projects it faithfully to $\mathbb{R}$: every dispersion-free state has $f \leq 2$ (the odd loop admits no two adjacent atoms valued 1), so $f(\overline{\mathrm{conv}}(S_{\mathrm{df}})) = [0,2]$. The non-classical state $w \equiv \frac{1}{2}$ has $f = \frac{5}{2}$, past the face $f = 2$, hence outside the hull — contextual. (Here f is a *chosen* separating functional for this w; by the separating-hyperplane theorem a state is contextual iff *some* linear functional sends it past the hull's image.)

Status. No witness has been constructed: the product $\prod_n \mathrm{MO}_2$ fails (its contextual states are only finitely additive — diffuse states on its central $P(\mathbb{N})$, which σ-additivity forces to concentrate on points, killing the contextuality, §5.2); Wright is finite; no other construction is known. On the negative side, the cell is now known to be *empty for every OML admitting a Polish Borel point-realisation* (Derr–Williamson, Remark 5.1). What remains genuinely open is therefore the *non-representable* case: a witness, if one exists, must admit no such realisation — which is precisely the absence of a σ-Loomis–Sikorski representation (§4.3), a descriptive-set-theory question.

Remark 5.4 (The compactness-failure analogy). The σ-essential requirement places the open problem alongside a known Boolean phenomenon. Over a Boolean algebra, countable additivity of a finitely additive charge is not first-order axiomatizable: an ultraproduct of σ-additive charges can be finitely additive but not σ-additive, so every finite test is passed while the global property fails [29]. The σ-essential contextual state is the non-distributive analogue, but with one sharpening the Boolean case obscures: the obstruction is *finitely detected but globally caused.* Each finite sub-OML is classically interpretable — its restriction lies in the full local hull $\overline{\mathrm{conv}}(S_{\mathrm{df}}(F))$ — yet contextuality is detected already at a single finite F, against the strictly smaller hull $\overline{\mathrm{conv}}(\{s|_F : s \in S^{\sigma}_{\mathrm{df}}\})$ of restrictions of the *global* σ-states. The gap between the two is what the countable whole supplies; it is not a property that emerges only in the limit (a vanishing-in-the-limit margin would be failure of σ-additivity itself, the segregated $\prod_n \mathrm{MO}_2$ case, not a witness).

5.2. **The combinatorial route.** One might seek the object among concrete σ-orthocomplete OMLs, on the grounds that a state's σ-additivity invokes $\bigvee_n a_n$ only for orthogonal families (Definition 2.5).

This route is settled, and does not produce the object. The property that would make σ-additivity genuinely exceed finite additivity on a concrete non-Boolean lattice is the interleaving condition

$$(\bigstar) \quad \exists\, p,\ \exists\,\{a_n\} \text{ infinite orthogonal}, \quad p \wedge a_n = 0\ (\forall n), \quad p \not\leq a_n\ (\forall n), \tag{2}$$

and $(\bigstar)$ is satisfied *trivially* by the plain product $\prod_n \mathrm{MO}_2$: take $a_n = a$ in coordinate n (zero elsewhere), and $p = \bigvee_n b_n$ where $b_n = b$ in coordinate n (zero elsewhere) — a genuine countable *orthogonal* join, so $p = (b, b, b, \dots)$ exists by σ-orthocompleteness. Then $p \wedge a_n = 0$ (coordinate n: $a \wedge b = 0$ in MO_2; elsewhere $a_n = 0$) and $p \not\leq a_n$ (coordinate n: $b \not\leq a^\perp$). Since $\prod_n \mathrm{MO}_2$ is concrete, σ-complete (a product of finite, hence complete, lattices), non-Boolean, and infinite, it satisfies $(\bigstar)$ and *every* combinatorial hypothesis one might impose — yet it is exactly the *segregated* object the open problem must exclude. Call an OML *segregated* when its non-distributivity is confined to a direct product (or central decomposition) of finite blocks, so that every dispersion-free state factors through the blocks and the infinitary structure carries no contextuality of its own. So the combinatorial axioms (σ-completeness, concreteness, $(\bigstar)$) are *not* the discriminator; the missing ingredient is *non-segregation*, a structural rather than a closure property.

Remark 5.5 (Absence of two-valued homomorphisms; the Boolean boundary). The universal closure above (a faithful set representation forces distributivity) already rules out the point-based route for every non-Boolean OML; $\prod_n \mathrm{MO}_2$ shows the failure concretely, at the level of points.

First, MO_2 admits *no* two-valued homomorphism. Each of its four separating two-valued states ($a \mapsto 0,\ b \mapsto 0$, and so on) fails subadditivity,

$$\omega(a \vee b) = \omega(\mathbf{1}) = 1\ >\ 0 = \omega(a) + \omega(b),$$

so none preserves $\vee$. The product inherits this: the diagonal copy $\{\mathbf{0}, \mathbf{1}, (a, a, \dots), (a^\perp, \dots), (b, b, \dots), (b^\perp, \dots)\}$ is a sub-MO_2, and any homomorphism of $\prod_n \mathrm{MO}_2$ would restrict to one on it, of which there are none. (This is special to MO_2 and its products: a generic non-Boolean OML such as $\mathbf{2} \times \mathrm{MO}_2$ *does* carry homomorphisms, e.g. a projection. The universal obstruction is the distributivity argument, not homomorphism-scarcity.) So $\prod_n \mathrm{MO}_2$ is concrete (Definition 2.7: order-determining two-valued *states*) and σ-complete, yet carries no two-valued homomorphism at all — the point-free target is forced, not engineered.

The same arithmetic records the sharp Boolean boundary. An OML is Boolean iff it admits a *unital* set of *subadditive* states — one giving each nonzero a value 1 for some such state (Pták–Pulmannová [35, Thm 1], for OMLs; the orthomodular-poset version fails). A concrete non-Boolean OML cannot have its separating two-valued states form such a set: some must fail subadditivity — above, all four of MO_2's do. The property that forces Boolean is therefore subadditivity, a finitary condition, not σ-additivity.

The lesson survives elaboration. Navara's construction [32] — once its stateless Greechie block (the source of non-concreteness) is replaced by MO_2 — yields a concrete, σ-orthocomplete, non-Boolean OML carrying $(\bigstar)$; yet it is $\prod_n \mathrm{MO}_2$ with scaffolding, segregated in the same way, with no σ-tribe representation. So $(\bigstar)$ and concreteness together are easy and do *not* reach the open problem: the missing ingredient is non-segregation, and that is what the witness must supply.

The known results constrain the open (σ-complete) problem from both sides without closing it: *no* impossibility theorem covers the concrete class (Remark 4.1), while the two *theorem-closed* routes to an engine — RDP and MacNeille (§4.3(i),(ii)) — are ruled out, leaving only the third, σ-Stone duality, which is itself open. What is required is not a new idea in the abstract but a representation in that gap: non-distributive enough to evade routes (i),(ii), yet σ-faithful where the known dualities are merely finitary.

Remark 5.6 (Why $L(H)$+Gleason does *not* already settle this — the (A)/(B) point-space equivocation). "Probability from relations, point-free" may *look* delivered by $L(H)$+Gleason. It is not: the appearance rests on conflating two point-spaces, and *realisation* is defined as concentration on the second. The two are (A) the Hilbert rays of H, which Gleason *requires* — the density ρ in $s = \operatorname{tr}(\rho\,\cdot)$ is built from them — and (B) the McDonald–Bimbó dual filters $P(A)$, which Gleason *removes*, but only by being a representation theorem. Gleason is thus the *most* point-ful result available (every lattice state corresponds to the point datum ρ). What $L(H)$ exhibits is a *separation*, not point-freeness: σ-additive measures exist on the lattice (Gleason, for normal states), yet on the dual $S_0(L(H))$ no state extends to a charge at all (Proposition 3.2), so a fortiori none concentrates on $P(A)$. It cannot serve as the existence proof required, so the point-free σ-additive structure of §5 is the only candidate engine.

5.3. **Two axes of obstruction.** The recurring confusion of this subject (Remarks 5.5, 5.6) is best held as a 2×2 table on two *independent* axes, which must not be collapsed into one.

- **Axis 1 — distributivity = commutativity.** One condition in two dialects: a family of projections commutes *iff* the lattice they generate is distributive. "Commutativity" is the operator word, "distributivity" the lattice word, for the *same* dividing line — not different sides.
- **Axis 2 — concreteness.** A non-Boolean lattice can be *point-rich in rays* (atoms = 1-dim subspaces) yet *point-free in valuations* (no two-valued homomorphisms, Kochen–Specker), or the reverse: the two senses of "point" come apart (the (A)/(B) equivocation, Rmk 5.6).

	distributive (commuting)	**non-distributive (non-commuting)**
concrete (valuations span)	classical: Boolean σ-algebra, Loomis–Sikorski free	the descent witness lives here: a concrete σ-complete non-Boolean OML ($\prod_n \mathrm{MO}_2$ lies here, but segregated)
non-concrete (KS: few valuations)	atomless measure algebra (valuation-poor)	$L(H)$: rays abundant, valuations KS-barred — the operator / e.s.a. world, the excluded pole

There is not one "lattice $\leftrightarrow$ space" dictionary but two, one per row: *Birkhoff–von Neumann* (bottom row) sends elements to closed subspaces and atoms to **rays**, with $a \vee b$ the closed span — rich in ray-points, KS-barred from valuation-points, hence non-concrete; *Stone / concrete-logic* (Gudder, top row) sends elements to sets of outcomes and atoms to points of Ω — rich in valuation-points, hence concrete. The operator question and the descent question share Axis 1 but sit in opposite rows: spectral projections *force* $L(H)$ (bottom-right), whereas the descent witness must be top-right. So the descent problem is the one cell no operator algebra can occupy — why a witness must be a pasting / countable colimit of concrete blocks, not a Hilbert-space construction, and why $L(H)$+Gleason cannot settle it. In this language the open problem (route (iii)) is exactly: *is there a logic $\leftrightarrow$ geometry dictionary for the top-right cell?* — and a σ-Loomis–Sikorski for concrete OMLs is that missing dictionary. Axes 1 and 2 are independent: non-distributivity does not entail non-concreteness, and a witness must be concrete and non-distributive simultaneously.

6. Directions

With the extension axis closed on $L(H)$ (Proposition 3.2), the combinatorial route shown to deliver only segregated objects (§5.2), and the question reduced to a σ-essential contextual state (§5.1), the problem admits two mutually exclusive resolutions: construct such a state, or prove none exists. The second is already established in the Polish-representable case (Derr–Williamson, Remark 5.1), so both turn on a single feature — whether a witness can fail to admit a regular Borel point-realisation.

Exhibit a σ-essential witness:

Construct an infinite concrete OML L carrying a σ-additive state w that is contextual yet witnessed by no finite sub-OML. The object can be pinned down. Write, for a finite sub-OML F,

$$A_F := \overline{\mathrm{conv}}\,(S_{\mathrm{df}}(F)), \qquad B_F := \overline{\mathrm{conv}}\,\{\, s|_F : s \in S^{\sigma}_{\mathrm{df}} \,\},$$

the full local dispersion-free hull and the hull of restrictions of the *global* σ-states. The inclusion

$$B_F \;\subseteq\; A_F$$

always holds, since a restriction $s|_F$ of a global dispersion-free state is a dispersion-free state on F. The two defining properties of w now combine. Because w is contextual, $w \notin \overline{\mathrm{conv}}(S^{\sigma}_{\mathrm{df}})$; this hull is compact, so by Hahn–Banach a finite-support functional separates w from it, and the elements in its support generate a finite sub-OML F with

$$w|_F \;\notin\; B_F.$$

Because w is σ-essential, on that same F

$$w|_F \;\in\; A_F.$$

The two statements, with $B_F \subseteq A_F$, give $w|_F \in A_F \setminus B_F$, so

$$B_F \;\subsetneq\; A_F.$$

The upshot: contextuality is finitely *detected* — it shows at the single finite F — though no finite sub-OML is itself contextual, and the whole target reduces to this one *restriction gap*, the failure of the global σ-states to restrict onto the full local hull. The obstruction is therefore carried by the σ-structure, not by any finite block.

One necessary condition follows. The witness must be *off-centre* (irreducible, with non-central infinite blocks): a central decomposition would route the infinitary content into a Boolean centre, where σ-additivity concentrates the contextual states onto points and removes them — this is the $\prod_n \mathrm{MO}_2$ mechanism (§5.2). The two obvious recipes both fail this test: a compactness-failure limit (Remark 5.4) collapses to non-σ-additivity, and a pasting or countable colimit of Wright-type blocks collapses to segregation.

Prove no such witness exists:

A spanning theorem: the restriction gap above is always empty (off a contextual finite sub-OML, Wright), so σ-essential relational probability is operationally invisible — present finitely or not at all. This is already established in the Polish-representable case (Derr–Williamson, Remark 5.1); a full result must extend it to the non-representable case, where their Theorem D.6 hypothesis is unavailable. The Boolean boundary (subadditivity of a unital set of states, Remark 5.5) is a natural lever, but cannot reuse the $\prod_n \mathrm{MO}_2$ mechanism, vacuous off-centre.

This also settles a subsidiary question: whether a singular state might extend where no normal state does, which would give a first example of extension without concentration. Proposition 3.2 excludes it by a *finite* family ($n = 4$), so the singular case — which was expected to require infinitary methods — is a self-contained finitary matter, not governed by the finitary-to-σ passage that carries the open question.

Appendix A. Glossary of definitions

A self-contained list of the notions used in the paper. Entries that restate a numbered definition in the body cross-reference it; the remaining entries are stated here. Throughout, L is an orthomodular lattice with orthocomplementation $a \mapsto a^{\perp}$, bottom $\mathbf{0}$, top $\mathbf{1}$.

Lattice notions.

Orthocomplemented lattice; orthomodular lattice (OML).: A bounded lattice with an order-reversing, involutive complement $a^\perp$ satisfying $a \vee a^\perp = \mathbf{1}$, $a \wedge a^\perp = \mathbf{0}$ (Definition 2.1); it is *orthomodular* if $a \le b \Rightarrow b = a \vee (b \wedge a^\perp)$ (Definition 2.4). A *Boolean algebra* is a distributive OML.

Orthogonal; meet-zero; incomparable.: $a \perp b$ means $a \le b^\perp$; *meet-zero* means $a \wedge b = \mathbf{0}$. Orthogonal implies meet-zero; the converse fails off the Boolean case, and that gap drives the subject (Definition 2.2).

Compatible; block.: a, b are *compatible* if they generate a Boolean subalgebra, equivalently $a = (a \wedge b) \vee (a \wedge b^\perp)$. A *block* is a maximal pairwise-compatible subset of L; it is a maximal Boolean sub-OML, and every OML is the union of its blocks. (In a concrete logic, compatibility of A, B is $A = (A \cap B) \cup (A \cap B^c)$.)

Completeness.: L is *σ-complete* if every countable subset has a join; it is *σ-orthocomplete* if every countable *pairwise-orthogonal* family has a join (Definition 2.5).

Centre; irreducible.: The *centre* $Z(L)$ is the set of elements compatible with all of L; it is a Boolean subalgebra. L is *irreducible* if $Z(L) = \{\mathbf{0}, \mathbf{1}\}$ (no nontrivial direct-product decomposition).

MO_n; $L(H)$.: MO_n is the modular ortholattice with $\mathbf{0}, \mathbf{1}$ and n complementary atom-pairs (MO_2: atoms $a, a^\perp, b, b^\perp$, all distinct meets $\mathbf{0}$); it is non-distributive for $n \ge 2$. $L(H)$ is the lattice of closed subspaces of a Hilbert space H (orthocomplement = orthogonal complement, join = closed span), the prototype non-distributive OML.

Representations.

OMP; concrete logic; order-determining states.: An *orthomodular poset* (OMP) weakens an OML to a partial join (only of orthogonal elements). L is a *concrete logic* (set-representable) if it embeds in some $\mathcal{P}(\Omega)$ with $a^\perp \mapsto \Omega \setminus a$ and orthogonal joins as disjoint unions, equivalently if it carries an *order-determining* (separating) family of two-valued states (Definition 2.7).

Faithful set representation; σ-tribe.: An injection $h : L \to \mathcal{P}(\Omega)$ with $h(a \wedge b) = h(a) \cap h(b)$, $h(a \vee b) = h(a) \cup h(b)$, $h(a^\perp) = \Omega \setminus h(a)$ (preserving *all* meets and joins as $\cap, \cup$); a *σ-tribe* / Loomis–Sikorski representation also preserves countable joins. By Theorem 5.2 only Boolean L admits one. (Contrast a concrete logic, which preserves only *orthogonal* joins as unions.)

Dual space $S_0(A)$; canonical points $P(A)$.: $S_0(A)$ is the McDonald–Bimbó space of filters of A; $P(A) = \{\uparrow p : p \in A\}$ is the set of principal filters, one per element (Definition 2.12).

States and measures.

State; meet-additive; charge.: A *state* is $s : L \to [0, 1]$ with $s(\mathbf{1}) = 1$ and $s(a \vee b) = s(a) + s(b)$ for $a \perp b$ (Definition 2.9); a *charge* is a finitely additive set function on a Boolean algebra (Definition 2.13).

σ-additive state.: A state s with $s\big(\bigvee_n a_n\big) = \sum_n s(a_n)$ for every countable pairwise-orthogonal $\{a_n\}$ (Definition 2.5).

Dispersion-free state.: A two-valued state, i.e. a state taking only the values $0, 1$ (equivalently a $\{0, 1\}$-homomorphism on each block); S_{df} denotes their set and S_{df}^{σ} the σ-additive ones. These are the deterministic / hidden-variable points.

Normal; singular state on $L(H)$.:
s is *normal* if $s(\cdot) = \mathrm{tr}(\rho\,\cdot)$ for a density operator ρ; *singular* if s vanishes on every finite-rank projection (Definition 2.11).

Subadditive state.:
A state s with $s(a \vee b) \leq s(a) + s(b)$ for *all* a, b (not only orthogonal pairs). A family of states is *unital* if for each $a \neq \mathbf{0}$ some member gives $s(a) = 1$.

The descent vocabulary.

Contextual state.:
A state w that is *not* in the closed convex hull $\overline{\mathrm{conv}}(S_{\mathrm{df}})$ of the dispersion-free states — equivalently (Lemma 5.3) a state with no hidden realisation, i.e. not a mixture of dispersion-free states. A non-contextual state is one that is such a mixture.

σ-essentially contextual state.:
A σ-additive state w on a σ-complete L that is *contextual* — $w \notin \overline{\mathrm{conv}}(S_{\mathrm{df}}^{\sigma})$ — yet whose restriction to *every finite sub-OML* of L is non-contextual. (All three clauses are required: σ-additive, globally contextual, finitely non-contextual. The finite version without the third clause is already inhabited, Wright's pentagon.)

Extension axis; descent axis.:
The *extension* axis asks whether a state extends to a charge on the full clopen algebra of $S_0(A)$ (a Horn–Tarski feasibility question); the *descent* axis asks whether such a charge concentrates on the physical points $P(A)$ (governed by σ-additivity). See §3.

Segregated.:
Of an OML: one whose non-distributivity is confined to finite blocks, so that $S_{\mathrm{df}} = S_{\mathrm{df}}^{\sigma}$ and no σ-essential leak is possible (e.g. $\prod_n \mathrm{MO}_2$, §5.2).

Prior-art objects (§5, Remark 5.1).

Pre-Dynkin-system; Dynkin-system.:
In Derr–Williamson's terminology, a *pre-Dynkin-system* on Ω is a concrete OMP (a family of subsets containing Ω, closed under complement and under disjoint unions); a *Dynkin-system* additionally is closed under countable disjoint unions — a concrete σ-complete OML.

σ-extendable.:
In Derr–Williamson, a countably additive probability μ_σ on a Dynkin-system $D_\sigma \subseteq \mathrm{Borel}(\Omega)$ (Ω Polish) is *σ-extendable* if it extends to a countably additive probability on $\mathrm{Borel}(\Omega)$.

Polish space; inner regular.:
A *Polish* space is a separable completely metrizable space. A measure μ is *inner regular* if $\mu(F) = \sup\{\mu(K) : K \subseteq F,\ K \text{ compact}\}$.

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